SHIVANI SHAH

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EDUCATION

University of Michigan — **Ann Arbor, MI**
BSE, Electrical Engineering - **GPA:** 3.606/4.000 — April 2017
MSE, Embedded Systems - **GPA:** 3.218/4.000 — April 2018

WORK EXPERIENCE

Limitless Dynamics — **Los Gatos, CA**
Electrical Engineering Contractor — February 2025 - March 2025
- Specced actuators for a robotic exoskeleton, evaluating torque density, power requirements, firmware integration and mechanical integration for human-assistive applications.

Tesla Energy — **Palo Alto, CA**
Staff Software Development Engineer – Test Infrastructure — August 2021 - October 2025
- Designed and developed Processor-in-the-Loop (PIL) test systems for energy products (Powerwall 3, Superchargers V3/V4, Megapack 2, Meter Socket Adapter), integrating custom Verilog for FPGA-based circuit emulation, Python for automated testing, and custom printed circuit board and mechanical CAD.
- Designed 15+ low-voltage, mixed-signal PCBs for Processor-in-the-Loop testers, integrating precision analog circuitry with high-speed digital interfaces such as LVDS and Gigabit Ethernet, ensuring reliable signal integrity and robust system performance.
- Led a team of 2 engineers in the development and deployment of PIL testers, with recent designs supporting early prototypes of next-gen Megapacks and validating firmware prior to system integration.

Neuralink — **San Francisco, CA**
System Electronics – Surgical Robot — June 2018 - July 2021
- Designed an Illumination Driver for fifteen LEDs to pulse with independent duty cycles synchronously with robot camera frame rates, reducing power consumption per LED and enabling frame interleaving of cameras
- Designed a printed circuit board that integrates all peripherals required for robot digital input and output including hit and hold solenoid drivers and closed loop laser drivers reducing build times by multiple days
- Debugged signal integrity of an LVDS commutator used for multiple hours of continuous rat brain recordings

Tesla Energy — **Palo Alto, CA**
Firmware Intern — Summer 2017
- Determined analog circuit and firmware parameters for new arc fault detection through data collection and simulation in Matlab

LEADERSHIP EXPERIENCE

Baja Racing SAE — **Ann Arbor, MI**
CVT System Co-Lead, Testing Lead — September 2013-April 2018
- Worked with a group to lead the design of Michigan Baja's first ever electronically controlled variable transmission (EVT) which has the potential to reduce tuning and manufacturing time of the system.
- Utilized strain gauges and hall effect sensors on a custom CVT dyno to measure torque and speed of both primary and secondary clutches. This data allowed us to verify efficiency of CVT and shifting characteristics such as engagement point, steady state RPM of the primary clutch and shift ratio.

Embedded Control Systems — **Ann Arbor, MI**
Graduate Instructional Assistant — January 2017-April 2018
- Lead weekly labs and office hours. Course covers and implements topics including memory mapped I/O, Interrupt Basics, CAN communication, PID control, and Real-time Scheduling

VOLUNTEERING

SF Ed Fund — **San Francisco, CA**
5th Grade Math Teacher's Assistant — March 2025-Present

SKILLS

Computer Aided Design: Solidworks (Mechanical And Electrical 2D), CATIA, Altium, Cadence (Schematic)
Communication Protocols: I2C, SPI, UART, CAN, LVDS, Ethernet
Coding Languages: C, C++, Python, Verilog, Git